

PRACTICAL - 5

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Section: 8

Experiment No. 5

Title: Implementation of a Producer-Consumer Communication System and a Shell Pipeline Emulator using Anonymous Pipes, fork(), dup2(), and exec() System Calls.

Aim:

To develop C programs that (i) implement a producer-consumer system using anonymous pipes where a parent process produces data and a child process consumes it, measuring the communication efficiency, and (ii) emulate the shell command `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

Objective:

- To understand inter-process communication (IPC) using anonymous pipes.
- To implement a producer-consumer relationship between a parent and child process.
- To measure and evaluate the communication efficiency (throughput) of pipe-based IPC.
- To understand file descriptor duplication using `dup2()`.
- To emulate shell command piping between two processes using `pipe()`, `fork()`, `dup2()`, and `exec()`.

Theory:

An anonymous pipe is a unidirectional inter-process communication mechanism provided by the operating system that allows related processes (such as a parent and its child created via `fork()`) to exchange data. A pipe has two ends represented by file descriptors: `fd[0]` for reading and `fd[1]` for writing. Data written to the write end can be read sequentially from the read end, and the pipe acts as a bounded, in-kernel buffer.

In a producer-consumer system, the producer (parent process) writes data into the pipe, while the consumer (child process) reads and processes this data. Since the pipe has a limited buffer size, the OS automatically blocks the producer when the pipe is full and blocks the consumer when the pipe is empty, providing implicit synchronization.

The `dup2()` system call is used to duplicate a file descriptor onto another descriptor number, commonly used to redirect standard input/output to a pipe. This is exactly how the shell implements command pipelines such as `ls -l | grep ".c"`: the standard output of the first command is redirected to the write end of a pipe, and the standard input of the second command is redirected to the read end of the same pipe, after which each command is executed using the `exec()` family of system calls.

Communication efficiency in this context is measured as throughput — the amount of data transferred through the pipe per unit time — calculated by timing the total duration of the write operations against the total volume of data transferred.

Algorithm (Part 1 – Producer-Consumer):

- Start the program and create a pipe using `pipe()`.
- Create a child process using `fork()`.

- In the child (consumer): close the unused write end, read data from the pipe in a loop until EOF, and record the total bytes received.
- In the parent (producer): close the unused read end, note the start time, write a fixed number of data chunks into the pipe, close the write end to signal EOF, and wait for the child.
- Note the end time in the parent and compute elapsed time and throughput.
- Display the number of bytes transferred and the communication efficiency.
- Stop.

Algorithm (Part 2 – ls -l | grep ".c" Emulation):

- Start the program and create a pipe using pipe().
- Create the first child process using fork().
- In the first child: redirect stdout to the pipe's write end using dup2(), close both pipe descriptors, and execute execlp("ls", "ls", "-l", NULL).
- Create the second child process using fork().
- In the second child: redirect stdin to the pipe's read end using dup2(), close both pipe descriptors, and execute execlp("grep", "grep", ".c", NULL).
- In the parent: close both ends of the pipe (so no descriptor keeps it open) and wait for both children using waitpid().
- Display a completion message.
- Stop.

System Calls Used:

System Call	Purpose
pipe()	Creates a unidirectional communication channel (read and write file descriptors)
fork()	Creates a child process
dup2()	Duplicates a file descriptor onto stdin/stdout for redirection
exec() family	Replaces a process image to run an external command (ls, grep)
read() / write()	Transfers data through the pipe
wait() / waitpid()	Synchronizes parent and child process completion
close()	Closes unused pipe file descriptors

Program 1: Producer-Consumer with Communication Efficiency Measurement

File: producer_consumer.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>
#include <sys/time.h>

#define CHUNK_SIZE 64
#define NUM_CHUNKS 1000

int main() {
    int fd[2];
    pid_t pid;
    struct timeval start, end;
```

```

if (pipe(fd) < 0) {
    perror("pipe failed");
    exit(1);
}

pid = fork();

if (pid < 0) {
    perror("fork failed");
    exit(1);
}

if (pid == 0) {
    /* ---- Child Process : Consumer ---- */
    char buffer[CHUNK_SIZE];
    int total_bytes = 0;
    close(fd[1]); /* close write end, child only reads */

    int n;
    while ((n = read(fd[0], buffer, CHUNK_SIZE)) > 0) {
        total_bytes += n;
    }

    printf("\nChild (Consumer) Process\n");
    printf("Child PID      : %d\n", getpid());
    printf("Total bytes read: %d\n", total_bytes);

    close(fd[0]);
    exit(0);
} else {
    /* ---- Parent Process : Producer ---- */
    char data[CHUNK_SIZE];
    memset(data, 'A', CHUNK_SIZE);
    close(fd[0]); /* close read end, parent only writes */

    gettimeofday(&start, NULL);
    for (int i = 0; i < NUM_CHUNKS; i++) {
        write(fd[1], data, CHUNK_SIZE);
    }
    close(fd[1]); /* signals EOF to child */

    wait(NULL);
    gettimeofday(&end, NULL);

    double elapsed = (end.tv_sec - start.tv_sec) +
                     (end.tv_usec - start.tv_usec) / 1e6;
    long total_sent = (long)CHUNK_SIZE * NUM_CHUNKS;
    double throughput = total_sent / elapsed / 1024.0; /* KB/s */

    printf("\nParent (Producer) Process\n");
    printf("Parent PID      : %d\n", getpid());
    printf("Total bytes sent : %ld\n", total_sent);
    printf("Time taken       : %f seconds\n", elapsed);
    printf("Communication efficiency (throughput): %.2f KB/s\n", throughput);
}
return 0;
}

```

Sample Output (Program 1):

```
teja@Hasini:~$ nano producer_consumer.c
teja@Hasini:~$ gcc producer_consumer.c -o producer_consumer
teja@Hasini:~$ ./producer_consumer

Child (Consumer) Process
Child PID      : 924
Total bytes read : 64000

Parent (Producer) Process
Parent PID     : 923
Total bytes sent : 64000
Time taken     : 0.000930 seconds
Communication efficiency (throughput): 67204.30 KB/s
teja@Hasini:~$ wsl
Command 'wsl' not found, but can be installed with:
sudo apt install wsl
teja@Hasini:~$ ./producer_consumer

Child (Consumer) Process
Child PID      : 953
Total bytes read : 64000

Parent (Producer) Process
Parent PID     : 952
Total bytes sent : 64000
Time taken     : 0.000461 seconds
Communication efficiency (throughput): 135574.84 KB/s
teja@Hasini:~$ ./producer_consumer

Child (Consumer) Process
Child PID      : 964
Total bytes read : 64000

Parent (Producer) Process
Parent PID     : 963
Total bytes sent : 64000
Time taken     : 0.000673 seconds
```

Program 2: Emulating ls -l | grep ".c" using pipe(), fork(), dup2(), exec()

File: pipeline.c

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
```

```

int main() {
    int fd[2];
    pid_t pid1, pid2;

    if (pipe(fd) < 0) {
        perror("pipe failed");
        exit(1);
    }

    /* ---- First child : executes ls -l ---- */
    pid1 = fork();
    if (pid1 < 0) {
        perror("fork failed");
        exit(1);
    }
    if (pid1 == 0) {
        dup2(fd[1], STDOUT_FILENO); /* redirect stdout to pipe write end */
        close(fd[0]);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
        perror("exec ls failed");
        exit(1);
    }

    /* ---- Second child : executes grep ".c" ---- */
    pid2 = fork();
    if (pid2 < 0) {
        perror("fork failed");
        exit(1);
    }
    if (pid2 == 0) {
        dup2(fd[0], STDIN_FILENO); /* redirect stdin from pipe read end */
        close(fd[0]);
        close(fd[1]);
        execlp("grep", "grep", ".c", NULL);
        perror("exec grep failed");
        exit(1);
    }

    /* ---- Parent : closes both ends and waits ---- */
    close(fd[0]);
    close(fd[1]);
    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);

    printf("\nPipeline execution completed (ls -l | grep \".c\").\n");
    return 0;
}

```

Sample Output (Program 2):

```

teja@Hasini:~$ nano pipeline.c
teja@Hasini:~$ nano pipeline.c
teja@Hasini:~$ gcc pipeline.c -o pipeline
teja@Hasini:~$ ./pipeline
-rw----- 1 teja teja 1034 Aug  8 10:06 filecopy.c.save
-rw----- 1 teja teja 1180 Aug  8 10:07 filecopy.c.save.1
-rw-r--r-- 1 teja teja  471 Aug 18 06:04 fork.c
-rw-r--r-- 1 teja teja 1180 Aug  8 10:08 hello.c
-rw-r--r-- 1 teja teja  896 Jul 30 04:56 os_task.c
-rw-r--r-- 1 teja teja 3258 Aug  8 10:00 parser.c
-rw-r--r-- 1 teja teja 2595 Aug 27 05:44 path_resolver.c
-rw-r--r-- 1 teja teja 1218 Sep 17 05:03 pipeline.c
-rwxr-xr-x 1 teja teja 16528 Sep 17 05:01 producer_consumer
-rw-r--r-- 1 teja teja 2084 Sep 17 05:01 producer_consumer.c
-rw----- 1 teja teja 2084 Sep 17 05:00 producer_consumer.c.sa
ve
-rw----- 1 teja teja 2084 Sep 17 05:01 producer_consumer.c.sa
ve.1
-rw-r--r-- 1 teja teja 7352 Sep 17 03:31 question.c
-rw-r--r-- 1 teja teja 2181 Sep 17 03:13 question7.c
-rw-r--r-- 1 teja teja 4530 Sep 17 03:21 question9.c
-rw-r--r-- 1 teja teja 3576 Aug 25 03:49 quotes.c
-rwxr-xr-x 1 teja teja 21632 Sep 17 03:42 redirection
-rw-r--r-- 1 teja teja 7814 Sep 17 03:42 redirection.c
-rwxr-xr-x 1 teja teja 21696 Sep 17 03:51 redirection_pipeline
-rw-r--r-- 1 teja teja 18366 Sep 17 03:51 redirection_pipeline.c

```

Pipeline execution completed (ls -l | grep ".c").

```

teja@Hasini:~$ touch abc.c
touch program.c
touch hello.txt
touch sample.py
teja@Hasini:~$ ./pipeline
-rw-r--r-- 1 teja teja    0 Sep 17 05:04 abc.c
-rw----- 1 teja teja 1034 Aug  8 10:06 filecopy.c.save
-rw----- 1 teja teja 1180 Aug  8 10:07 filecopy.c.save.1
-rw-r--r-- 1 teja teja  471 Aug 18 06:04 fork.c

```

```

teja@Hasini:~$ ./pipeline
-rw-r--r-- 1 teja teja      0 Sep 17 05:04 abc.c
-rw----- 1 teja teja  1034 Aug  8 10:06 filecopy.c.save
-rw----- 1 teja teja  1180 Aug  8 10:07 filecopy.c.save.1
-rw-r--r-- 1 teja teja   471 Aug 18 06:04 fork.c
-rw-r--r-- 1 teja teja  1180 Aug  8 10:08 hello.c
-rw-r--r-- 1 teja teja   896 Jul 30 04:56 os_task.c
-rw-r--r-- 1 teja teja  3258 Aug  8 10:00 parser.c
-rw-r--r-- 1 teja teja  2595 Aug 27 05:44 path_resolver.c
-rw-r--r-- 1 teja teja  1218 Sep 17 05:03 pipeline.c
-rwxr-xr-x 1 teja teja 16528 Sep 17 05:01 producer_consumer
-rw-r--r-- 1 teja teja  2084 Sep 17 05:01 producer_consumer.c
-rw----- 1 teja teja  2084 Sep 17 05:00 producer_consumer.c.sa
ve
-rw----- 1 teja teja  2084 Sep 17 05:01 producer_consumer.c.sa
ve.1
-rw-r--r-- 1 teja teja      0 Sep 17 05:04 program.c
-rw-r--r-- 1 teja teja  7352 Sep 17 03:31 question.c
-rw-r--r-- 1 teja teja  2181 Sep 17 03:13 question7.c
-rw-r--r-- 1 teja teja  4530 Sep 17 03:21 question9.c
-rw-r--r-- 1 teja teja  3576 Aug 25 03:49 quotes.c
-rwxr-xr-x 1 teja teja 21632 Sep 17 03:42 redirection
-rw-r--r-- 1 teja teja  7814 Sep 17 03:42 redirection.c
-rwxr-xr-x 1 teja teja 21696 Sep 17 03:51 redirection_pipeline
-rw-r--r-- 1 teja teja 18366 Sep 17 03:51 redirection_pipeline.c

Pipeline execution completed (ls -l | grep ".c").
teja@Hasini:~$ ls -l | grep ".c"
-rw-r--r-- 1 teja teja      0 Sep 17 05:04 abc.c
-rw----- 1 teja teja  1034 Aug  8 10:06 filecopy.c.save
-rw----- 1 teja teja  1180 Aug  8 10:07 filecopy.c.save.1
-rw-r--r-- 1 teja teja   471 Aug 18 06:04 fork.c
-rw-r--r-- 1 teja teja  1180 Aug  8 10:08 hello.c
-rw-r--r-- 1 teja teja   896 Jul 30 04:56 os_task.c
-rw-r--r-- 1 teja teja  3258 Aug  8 10:00 parser.c

```

Observation:

- A pipe was successfully created and used to transfer data between a parent (producer) and child (consumer) process.
- The operating system's pipe buffer provided implicit synchronization between the producer and consumer without needing explicit locks.
- The communication efficiency (throughput) was computed by timing the write operations against the volume of data transferred through the pipe.
- Using `dup2()`, standard output of one process and standard input of another were successfully redirected to the two ends of a pipe, allowing `exec()`-launched programs (`ls` and `grep`) to communicate exactly as the shell pipeline `ls -l | grep ".c"` would.
- Closing unused pipe ends in every process (including the parent) was essential — otherwise the reading process never receives EOF and blocks indefinitely.

Result:

The programs were successfully executed. A producer-consumer communication system was implemented using anonymous pipes, and the communication efficiency (throughput) was measured. The shell command `ls -l | grep ".c"` was successfully emulated using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls, correctly filtering and displaying only the ".c" files from the directory listing.